Bit Accuracy Hides Prototype Collapse in Hebbian Associative Memory

An integration-depth analysis of phase-coupled and Hopfield retrieval of correlated patterns

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Code:

https://github.com/labinot-marku/CONN-Associative-Memory/tree/main/similarity-discrimination

Abstract

When stored patterns are highly correlated, standard recall metrics can report success while the tested decoders can no longer identify the original stored pattern from the network's output. We study this with families of correlated binary patterns (70–95% within-family similarity) and ask whether a retrieval system can identify the specific family member from a noisy cue. A converged Hopfield network reaches 88% bit-level recall but identifies the correct member at chance (26%) and yields negligible nearest-neighbour decoder-level mutual information (≤ 0.05 bits; nominally significant in one of five nearest-neighbour tests, $p = 0.03$, and not in the other four). A minimally evolving phase-coupled system preserves the member identity already present in the noisy cue: it achieves 94% discrimination and 1.70 bits of decoder-level mutual information (95% CI 1.56–1.88, seed-clustered; maximum 2.0), despite lower bit accuracy (75%). We call this the *bit-accuracy inversion*. It persists when the family prototype is not stored and when all stored family members are pairwise equidistant; in the equidistant design, converged Hopfield reaches 91.7% bit accuracy while the tested decoders detect no member information, and its binarised outputs coincide exactly with the unstored family prototype in 88–90% of trials.

Matched controls then locate the cause and show that the phase system does not perform associative retrieval. The phase network at its published parameters moves 0.2% of bits; the raw noisy cue with no dynamics gives the same result, and random or zero coupling matrices give the same result as the learned Hebbian one. Hopfield networks run for only a few steps preserve member identity just as well: a continuous Hopfield network after 5 Euler steps scores 94.2%, and an asynchronous binary network after 8 single-unit updates scores 89.6%. Across the 31 tested configurations of phase, field-coupled, continuous and binary Hopfield dynamics, discrimination follows a common decreasing trend with how far the output has moved from the cue. The discrimination advantage reported for phase dynamics in earlier versions is therefore explained by their limited integration depth, not by a demonstrated advantage of phase coupling or learned connectivity. Converged Hebbian dynamics drive correlated patterns toward the family prototype or nearby prototype-like states. A trained decoder tested on held-out seeds finds no more member information than the nearest-neighbour readout, and a decoder-independent bound shows that, under matched noise draws, the binarised converged outputs are identical for all four possible cues in 56–96% of sampled cases, limiting the member information they can carry to an estimated 0.08–0.88 bits (seed-level upper 95% limits 0.15–1.07 bits). The bound concerns the binarised output; it does not extend to small residual differences in the continuous network state. Bit-level accuracy cannot detect this loss; decoder-level member information can. All code, per-trial predictions and parameter sweeps are public.

Keywords: associative memory, Hopfield network, correlated patterns, prototype attractor, mutual information, oscillatory networks, evaluation metrics

1 Introduction

Associative memory networks store patterns as attractors of a dynamical system and retrieve them by relaxation from a noisy cue [1]. For random patterns the classical Hopfield network stores about $0.138N$ patterns [2]. Correlated patterns behave differently. With Hebbian learning, a family of patterns that share most of their components produces an attractor near the family centroid, so the network retrieves the category rather than the individual member [3, 4, 6, 7, 8]. In dense associative memories, Krotov and Hopfield described a transition from a feature-matching to a prototype regime [5], and the related transition from memorisation to prototype formation has recently been described as the onset of generalisation [9].

This paper is about how that collapse is measured. If the prototype is 85% identical to each family member, a network that always returns the prototype scores 85% bit accuracy on each non-prototype member (88.75% averaged over a family in which the prototype is itself stored) while carrying no information about which member was cued. Bit accuracy therefore rewards exactly the failure one wants to detect. We quantify the effect with a direct measure: the decoder-level mutual information between the cued member and the member identified from the network output.

The work began as a test of whether phase-coupled oscillator networks [10, 11, 12, 13], as in the CONN model [14], discriminate correlated patterns better than Hopfield networks. Earlier versions (V1–V4) reported a large advantage for phase coupling. Successive controls, reported here in the order they were run, show that the advantage comes from how far each system's dynamics move the state, not from its coupling. Version 5 added a direct test of Hopfield networks at controlled integration depth (Experiment 13) and corrected a bias-correction error in the mutual-information estimator. Version 6 tests whether the results depend on how the pattern families are constructed (Experiment 15) and treats seeds as the unit of statistical inference. Version 7 measures the outputs' distances to the prototype directly and tests whether a trained decoder, or any decoder, could recover member information that the nearest-neighbour readout misses (Experiments 15b and 15c).

2 Methods

Phase-coupled network. Each unit j is a phase oscillator $\varphi_j \in [0, 2\pi)$. Patterns are binary phase configurations, $\varphi_j \in \{0, \pi\}$. The Hebbian matrix is $J_{ij} = \frac{1}{N}\sum_\mu \cos\varphi_i^\mu \cos\varphi_j^\mu$, $J_{ii} = 0$. The minimal system is

$$\varphi_i \leftarrow \varphi_i + \eta \sum_j J_{ij}\, f(\varphi_j - \varphi_i),$$

with $f = \sin$ unless stated, $\eta = 0.005$ and 150 steps (the "published parameters"). At these parameters the mean phase drift is about 10° and fewer than 0.2% of bits change sign. The output is read as $\mathrm{sign}(\cos\varphi_j)$.

Hopfield baselines. Binary Hopfield: synchronous updates $s \leftarrow \mathrm{sign}(Ws)$ to convergence (at most 100 steps), $W = \frac{1}{N}\sum_\mu \xi^\mu \xi^{\mu\top}$, $W_{ii} = 0$, with $\mathrm{sign}(0) = +1$. Experiments 1–14 stop at a fixed point or after 100 steps; Experiment 15 also stops when the newly computed state equals the state two updates earlier (a two-cycle) and returns that newly computed state. Asynchronous updates (Experiment 13) pick units uniformly at random with replacement and set $s_i = +1$ if $(Ws)_i \geq 0$, else -1. All methods start from the same noisy cue in each trial. Continuous Hopfield: $\dot{x} = -x + \tanh(\beta W x)$, $\beta = 2$, Euler step $dt = 0.1$, 150 steps (15 time constants, i.e. converged), started from $x(0) =$ the ± 1 cue plus Gaussian jitter ($\sigma = 0.15$), read out as $\mathrm{sign}(x)$. Experiment 13 varies the number of steps and adds asynchronous single-unit updates.

Pattern families. Unless stated: $N = 80$, 3 families of 4 members, 85% within-family similarity, 5 unrelated patterns, 17 patterns in total ($\alpha = 0.212$). Each family is built from a random prototype; member 0 is the prototype itself, members 1–3 differ from it in $(1 - s)N$ random positions. This gives a star-shaped family: members 1–3 are 85% similar to the prototype but only about 74.5% similar to each other ($1 - 2p(1 - p)$ with $p = 0.15$). Experiment 15 compares two balanced designs. Cues are the target with 25% of bits flipped.

Noise model. From Experiment 5 on, all systems receive the same binary cue. Phase systems receive it as phases plus Gaussian jitter ($\sigma = 0.15$); the jitter contributes 0.0 percentage points to discrimination (tested). The jitter is not inert: for exact binary phases every difference $\varphi_j - \varphi_i$ is 0 or π, so $\sin(\varphi_j - \varphi_i) = 0$ and the dynamics do not move. The jitter starts continuous phase evolution and may shape sub-threshold trajectories that the binarised readout does not show.

Readout. *Discrimination*: the fraction of trials in which the member with the highest overlap with the binarised output is the cued member (chance 25% for 4 members). *Bitwise accuracy*: the mean fraction of output bits equal to the target (called "exact recall" in V1–V6; it is not exact pattern retrieval, which would require every bit to match). *Displacement*: the fraction of bits whose sign differs between output and cue.

Decoder-level mutual information. $I(\text{cued member}; \text{identified member})$ is estimated from the joint frequency table of all trials pooled across seeds, with the Miller–Madow correction applied to each entropy term [15, 16]:

$$\hat{I}_{\text{MM}} = \hat{I}_{\text{plug-in}} + \frac{(m_X - 1) + (m_Y - 1) - (m_{XY} - 1)}{2n \ln 2} \text{ bits},$$

where m counts non-empty cells and n is the number of trials. Plug-in mutual information is biased upward [17], so the correction is normally negative. Estimates are clipped to $[0, \log_2 4]$; the same clipped estimator is applied to the point estimate, to every bootstrap replicate and to every permutation statistic, always to trials pooled across seeds before correction. The identified member $D(Z, S)$ is a deterministic function of the output state Z and the stored family S; the nearest-neighbour decoder has no fitted parameters. By the data-processing inequality, $I(Y; D(Z, S)) \leq I(Y; Z, S)$, and because the cued member Y is drawn independently of the family, $I(Y; Z, S) = I(Y; Z \mid S)$. Decoder-level MI is therefore a lower bound on the member information in the output state given knowledge of the stored family. Versions 1–4 used an incorrect correction; see Appendix E.

Statistics. Trials generated from the same seed share pattern families and are not independent, so the seed is the unit of resampling. Confidence intervals use a hierarchical bootstrap (resample seeds, then trials within each drawn seed; $B = 2000$; 2.5th–97.5th percentile). Whether MI differs from zero is tested by permuting identified labels within each seed (2000 permutations). The two procedures answer different questions. The bootstrap interval describes the sampling uncertainty of the MI *estimate*, which remains positively biased at these sample sizes even after correction, so an interval excluding zero is not evidence of dependence. The permutation test is the test of independence; claims that a system retains member information rest on it. For the trained decoder of Experiment 15c the permutation is applied to the pooled held-out predictions only; the decoder is not refitted for each permutation, so this is a conditional permutation test given the fitted decoders. Comparisons across seeds use the Wilcoxon signed-rank test. Random numbers come from NumPy's default generator, base seed 42 unless stated.

3 Results

Each experiment addresses the strongest objection to the previous one. Objections were collected through adversarial review by several AI systems (Section 9).

3.1 Experiment 1: Initial observation

Full CONN dynamics [14] versus binary Hopfield; $N = 80$, 15 seeds.

Metric	CONN	Hopfield	Δ
Target bitwise accuracy	87.7%	88.1%	−0.4
Correct member identified	99.4%	25.6%	+73.9
Right family (any member > 80%)	98.3%	97.2%	+1.1

Table 1: Experiment 1. Both systems find the right family; only CONN identifies the member. Wilcoxon $p < 0.0001$. Noise models were not yet matched between systems here; matched noise from Experiment 5 on.

3.2 Experiment 2: Component ablation

Removing CONN's amplitude dynamics and coherence prior leaves discrimination unchanged (99.4%, 99.4%, 99.4%, 98.9%). Bare sinusoidal coupling $\varphi_i \leftarrow \varphi_i + \eta \sum_j J_{ij} \sin(\varphi_j - \varphi_i)$ suffices; all later experiments use it.

3.3 Experiment 3: Similarity sweep

Similarity	70%	75%	80%	85%	90%	95%
Phase-coupled	98.8	98.8	97.5	92.9	90.4	72.5
Binary Hopfield	35.4	29.2	28.3	26.7	26.2	25.0

Table 2: Discrimination (%) under matched noise. V1/V2 reported higher phase values under asymmetric noise.

3.4 Experiment 4: Computational budget

Ten independent Hopfield retrieval attempts per cue do not improve discrimination (25.0% throughout). Extra converged attempts do not help.

3.5 Experiment 5: Matched conditions and continuous Hopfield

Under identical cues (20 seeds): phase 92.5% $\pm$ 7.4; continuous Hopfield 25.8% $\pm$ 3.6; binary Hopfield 25.8% $\pm$ 3.6 (phase vs. each, $p < 0.0001$). Both Hopfield variants were run to convergence; Experiment 13 shows that this, not the nonlinearity, is why they agree.

3.6 Experiment 6: Coupling function

Model	Coupling	Discrimination	Displacement
A. Sinusoidal	$\sin(\varphi_j - \varphi_i)$	93.3%	$< 0.2\%$
B. Linear circular	$(\varphi_j - \varphi_i) \bmod 2\pi$	93.3%	$< 0.2\%$
C. Cosine	$\cos(\varphi_j - \varphi_i)$	92.5%	$< 0.2\%$
D. tanh field on circle	$\tanh(J\varphi)$, $\varphi \in [0, 2\pi)$	25.8%	$\approx 26\%$
E. Binary Hopfield	$\text{sign}(Wx)$	25.0%	$\approx 30\%$

Table 3: Experiment 6, 85% similarity. Model D used uncentred phases as tanh input and therefore moved far more bits than A–C at the same η. A centred Model D at matched depth gives 92.9% (Experiment 14).

A five-model similarity sweep (70–95%) gives the same ordering at every similarity: A–C from 99.6% to about 70%, D and E between 19% and 38%.

3.7 Experiment 7: Mutual information and the bit-accuracy inversion

$N = 80$, 85% similarity, 15 seeds, 180 trials per system.

Metric	Phase	Continuous Hopfield	Binary Hopfield
Discrimination	94.4%	26.7%	27.8%
Bitwise accuracy	75.4%	87.4%	88.2%
Discrimination, 95% CI	[89.4, 98.3]	[20.0, 33.9]	[21.7, 35.0]
Decoder-level MI (bits)	1.703	0.040	0.048
MI, 95% CI	[1.560, 1.881]	[0.009, 0.126]	[0.009, 0.123]
Seed-stratified permutation test	$p < 0.001$	$p = 0.25$	$p = 0.03$
MI / maximum (2.0 bits)	85.1%	2.0%	2.4%

Table 4: Experiment 7. Intervals from the hierarchical bootstrap. Per-seed MI: phase vs. binary and phase vs. continuous Hopfield, Wilcoxon $p = 0.0003$ each.

The bit-accuracy inversion. Hopfield has the higher bit accuracy because it converges to states close to the stored prototype, which is 85% identical to every other member: 96.7% of converged binary outputs are decoded as the prototype (member 0), and in Experiment 15 their median distance from it is 2 of 80 bits (Table 12). The 69.4%/30.6% split between prototype and mixture states reported in V4 is not reproduced

by the deposited code and is withdrawn. Its member-specific information is not distinguishable from zero for the continuous network. For the binary network it is nominally significant ($p = 0.03$) but negligible (0.048 of 2 bits), and it does not replicate: converged binary Hopfield gives $p = 0.36$ in Experiment 13 and $p = 0.48$, 1.0 and 1.0 in the three designs of Experiment 15. With five tests, one $p = 0.03$ is consistent with chance. The bootstrap intervals exclude zero because plug-in estimates from 180 trials are biased upward; the permutation test is the appropriate check. The phase system has lower bit accuracy and keeps 85% of the maximum member information.

For 94.4% accuracy with errors spread uniformly over the three wrong members, MI would be 1.60 bits. Uniform errors give the lowest MI for a given accuracy, so the observed 1.70 bits simply means the errors are concentrated on a few member pairs.

3.8 Experiment 8: Network size

N	M	α	Phase disc.	Hopfield disc.	Phase MI	Hopfield MI	Phase bit acc.	Hopfield bit acc.
80	16	0.200	91.7%	26.7%	1.56	0.03	74.8%	88.0%
128	18	0.141	99.4%	25.0%	1.97	0.00	75.2%	89.0%
192	21	0.109	100.0%	25.0%	2.00	0.00	74.9%	89.2%
256	24	0.094	100.0%	25.0%	2.00	0.00	75.0%	89.0%

Table 5: Experiment 8 (15 seeds; MI in bits, pooled). With fixed family structure, larger N means more bits separate members and lower load, so the cue itself becomes more informative.

3.9 Experiment 9: Load

Patterns	α	Phase disc.	Hopfield disc.	Phase MI (bits)	Hopfield MI (bits)
12	0.150	93.9%	25.6%	1.62	0.01
17	0.212	93.3%	26.7%	1.60	0.04
22	0.275	93.3%	25.6%	1.61	0.02
32	0.400	91.1%	27.8%	1.46	0.04
42	0.525	92.8%	25.6%	1.54	0.01
52	0.650	96.1%	26.7%	1.73	0.00

Table 6: Experiment 9, $N = 80$, 15 seeds. The $\alpha = 0.650$ value is the output of the deposited script; V4 reported 93.3% from a separate run.

Phase discrimination does not depend on load, as expected if the dynamics barely move the state: the coupling matrix, which is where load acts, has little effect (Experiment 14).

3.10 Experiment 10: Discrimination versus bitwise accuracy across similarity

Similarity	Phase disc.	Phase bit acc.	Hopfield disc.	Hopfield bit acc.	Phase MI (bits)
70%	98.3%	74.6%	37.2%	77.5%	1.89
80%	96.1%	74.7%	25.6%	85.4%	1.73
85%	92.8%	74.7%	26.1%	87.7%	1.58
90%	88.3%	74.9%	25.6%	92.9%	1.37
95%	72.8%	75.4%	25.6%	95.9%	0.84

Table 7: Experiment 10. As similarity rises, Hopfield's bit accuracy rises towards the similarity level while its discrimination stays at chance: at 95% similarity its 95.9% bitwise accuracy is prototype overlap. Phase bit accuracy stays near 75%, the accuracy of the 25%-noise cue.

3.11 Experiment 11: Sparse connectivity at $N = 1024$

With 25% random connectivity, Models A–C discriminate perfectly (100.0%), while Model D and binary Hopfield give 25.0% (trial-level SD 43.3%, i.e. Bernoulli at chance). Model D is the uncentred field model of Experiment 6, which moves many more bits; this is a depth contrast, not a coupling contrast.

3.12 Experiment 12: Modern Hopfield network

A modern Hopfield network [18], $x \leftarrow X\,\mathrm{softmax}(\beta X^\top x)$ for 50 steps with X the stored ± 1 patterns, unnormalised dot products, input the jittered cue and output $\mathrm{sign}(x)$ ($\beta = 1$–16), gives 92.5–92.9%, the same as a hard nearest-neighbour lookup (92.9%). The top-1 versus top-2 overlap gap averages 10.4, so softmax weights differ by a factor of about $e^{10.4} \approx 33{,}000$ at $\beta = 1$; 99.2–100% of outputs equal the nearest-neighbour output. In this regime (binary patterns, $\beta \geq 1$, large overlap gaps) the softmax saturates and the modern Hopfield network behaves as a nearest-neighbour lookup. At smaller β or with smaller overlap gaps it would average across stored patterns; that regime was not tested, and the result says nothing general about modern Hopfield networks.

3.13 Experiment 13: Hopfield at controlled integration depth (new in V5)

V4 argued that Hopfield's sign update is structurally unable to run gently. This experiment tests that claim directly ($N = 80$, 85% similarity, 20 seeds, 240 trials per condition).

Condition	Discrimination	MI (bits)	Displacement
Raw noisy cue (no dynamics)	94.2%	1.63	0.0%
Continuous Hopfield, 1 / 2 / 5 steps	94.2%	1.63	0.0–0.4%
Continuous Hopfield, 10 steps	35.4%	0.17	26.6%
Continuous Hopfield, 25 steps	28.7%	0.05	29.6%
Continuous Hopfield, 150 steps (published)	26.2%	0.01	30.5%
Asynchronous binary, 8 unit updates	89.6%	1.45	2.8%
Asynchronous binary, 20 unit updates	83.8%	1.25	6.8%
Asynchronous binary, 40 unit updates	67.9%	0.87	11.9%
Asynchronous binary, 80 unit updates (1 sweep)	45.0%	0.34	18.9%
Asynchronous binary, 800 unit updates	27.9%	0.02	30.5%
Synchronous binary, 1 step	38.3%	0.22	29.8%
Synchronous binary, converged (published)	27.1%	0.02	30.2%

Table 8: Experiment 13 (`conn_hopfield_depth.py`). Hopfield dynamics stopped early preserve member identity as well as the phase network does. Seed-clustered 95% CIs for discrimination: raw cue 90.0–97.5%; asynchronous, 8 updates 84.2–94.6%; converged continuous 21.2–32.5%; converged synchronous 21.7–33.3%. Seed-stratified permutation tests on MI: $p < 0.001$ for all conditions up to 80 asynchronous updates and for 10 continuous steps; $p = 0.009$ at 25 continuous steps (0.05 bits); $p = 0.76$, 0.34 and 0.36 for converged continuous, 800 asynchronous updates and converged synchronous.

A continuous Hopfield network stopped after 5 Euler steps is indistinguishable from the raw cue. A binary network updated asynchronously for 8 units keeps 89.6%. Even one synchronous step leaves 38.3%, above chance, because it flips about 30% of bits at once. The Hopfield failure in Experiments 1–12 is a consequence of running to convergence, not of the sign nonlinearity.

3.14 Experiment 14: Integration-depth sensitivity

Parameter sweep. $\eta \in \{0.0025, 0.005, 0.010\}$, steps $\in \{50, \ldots, 1200\}$, 10 seeds.

The published parameters lie at the edge of the preservation regime ($\eta \times \text{steps} = 0.75$).

Matched-depth baselines. 20 seeds.

Weight-matrix independence. At the published parameters, the learned Hebbian J, a random J (i.i.d. Gaussian entries, zero diagonal, rescaled to the Frobenius norm of the learned J), and $J = 0$ all give 95.4%

η \ steps	50	150	300	600	1200
0.0025	93.3	93.3	93.3	85.8	40.8
0.0050	93.3	**93.3***	85.8	40.8	27.5
0.0100	93.3	85.8	40.0	27.5	27.5

Table 9: Phase discrimination (%). *Published parameters. Results depend only on $\eta\times$steps. MI: 1.65 bits for $\eta\times\text{steps} \leq 0.75$ ($< 0.3\%$ of bits moved), 1.38 bits at 1.5 (8.6% moved), 0.29–0.31 bits at 3.0, 0.03–0.04 bits beyond.

Condition	Discrimination	MI (bits)	Displacement
Raw cue, nearest neighbour	92.9%	1.60	0.0%
Phase short ($\eta = 0.005$, 150 steps) [published]	92.9%	1.60	0.2%
Field short (centred, $\eta = 0.001$, 150 steps, $\beta = 0.5$)	92.9%	1.60	0.0%
Phase long ($\eta = 0.005$, 600 steps)	38.3%	0.20	26.1%
Field long (centred, $\eta = 0.005$, 600 steps, $\beta = 1$)	25.0%	0.00	50.9%
Hopfield, converged	26.7%	0.01	$\approx 30\%$

Table 10: Wilcoxon vs. converged Hopfield: raw cue, phase short, field short +66.2 pp ($p < 0.001$); phase long +11.7 pp ($p = 0.002$); field long −1.7 pp ($p = 0.95$). The Hopfield displacement is from Experiment 13.

(20 seeds). Random initial phases with the learned J give chance (24.6%). The discrimination signal is in the cue, not in J.

Family size. With 2, 4, 8, 16 and 32 members per family, phase discrimination is 94.2, 92.9, 86.5, 15.8 and 3.8%, while the raw cue gives 94.2, 92.9, 90.8, 91.0 and 85.9% (Hopfield 55.0, 26.7, 14.0, 6.7, 3.4%). At 16 members ($\alpha = 0.60$) even the gentle phase dynamics destroy the signal, so preservation has a load limit that the cue alone does not.

Below-capacity Hopfield and unrelated patterns. Converged Hopfield fails to discriminate even at $\alpha = 0.150$, so the failure is not saturation. On unrelated (uncorrelated) patterns, phase bitwise accuracy is 74.5% and Hopfield 55.3%.

3.15 Experiment 15: Family geometry (new in V6)

The star-shaped families of Experiments 1–14 give the prototype a privileged position, and a network that always returned it would score $(1+3\times0.85)/4 = 88.75\%$ bit accuracy, close to converged Hopfield's 88.2%. To test whether the inversion depends on this construction, we compare three designs ($N = 80$, 3 families of 4, 5 unrelated patterns, 25% cue noise, 20 seeds, 240 trials per condition; `conn_family_geometry.py`):

- **A, star (V5):** prototype stored as member 0; members 1–3 flip 15% of its bits.
- **B, hidden prototype:** four members, each flipping 15% of a prototype that is never stored; all members are equivalent (pairwise similarity $\approx 74.5\%$).
- **C, equidistant:** each member flips its own disjoint set of 6 bits of a hidden prototype, so every pair of members is exactly 85% similar.

A *prototype-only* baseline returns the family's prototype on every trial, regardless of the cue.

The inversion does not depend on the star construction. It is strongest in the equidistant design, where converged Hopfield reaches 91.7% bit accuracy while the nearest-neighbour readout detects no member information. In all three designs, converged Hopfield is within 2 points of the prototype-only output in bit accuracy and within 3 points of it in discrimination. The near-match to the prototype-only baseline is indirect evidence. Table 12 measures the outputs' Hamming distances directly (`conn_prototype_distance.py`).

The distances support prototype attraction, with a different strength in each design. In the equidistant design the converged state *is* the unstored prototype in 88–90% of trials and is closer to it than to any stored member in 98–99%. In the star design outputs lie a median of 2 bits from the stored prototype but match it exactly in only 12–13% of trials. In the hidden-prototype design, where the members are farther apart (about 74.5% pairwise), outputs approach the unstored prototype (median 7.5% of bits) and are closer to it than to any member in 70% of trials, but never reach it; we describe this as prototype-like retrieval rather than convergence onto the prototype. Synchronous two-cycles occurred in at most 9 of

Condition	A: star	B: hidden prototype	C: equidistant members (85%)
Raw cue: disc. / MI	93.8% / 1.65	96.2% / 1.74	89.2% / 1.35
Phase, published ($J = 0$ identical)	93.8% / 1.65	96.2% / 1.74	89.2% / 1.35
Continuous Hopfield, 5 steps	93.8% / 1.65	96.2% / 1.74	89.2% / 1.35
Asynchronous binary, 8 updates	88.8% / 1.44	93.8% / 1.56	86.7% / 1.27
Converged continuous: disc.	27.1%	26.7%	25.4%
bit accuracy	87.2%	83.3%	**91.7%**
MI (bits), p	0.02, 0.28	0.00, 1.0	0.00, 1.0
Converged synchronous: disc.	27.1%	27.9%	25.8%
bit accuracy	87.9%	83.9%	91.7%
MI (bits), p	0.02, 0.48	0.00, 1.0	0.00, 1.0
2-cycles (of 240 runs)	6	9	0
Prototype-only output: disc. / bit acc.	25.0% / 88.7%	25.0% / 85.0%	25.0% / 92.5%

Table 11: Experiment 15. Discrimination / decoder-level MI in bits. p: seed-stratified permutation test. Seed-clustered 95% CIs for discrimination are about ±5 points throughout (`family_geometry_results.csv`).

Design	Condition	d(out, proto.)	d(out, target)	d(out, nearest member)	= prototype	closer to proto.
A: star	raw cue	31.1%	25.6%	26.9%	0%	23%
	converged (cont. / sync.)	4.8 / 4.4%	12.5 / 12.1%	13.7 / 13.4%	12 / 13%	90 / 91%
B: hidden prototype	raw cue	32.9%	25.1%	25.1%	0%	1%
	converged (cont. / sync.)	10.1 / 9.0%	16.8 / 15.9%	13.1 / 12.2%	0 / 0%	71 / 70%
C: equidistant	raw cue	29.1%	25.2%	25.1%	0%	2%
	converged (cont. / sync.)	0.6 / 0.5%	7.8 / 7.7%	7.6 / 7.6%	88 / 90%	99 / 98%

Table 12: Experiment 15b: mean Hamming distance (% of bits) of outputs to the family prototype, the cued member and the nearest stored member other than the prototype; share of outputs identical to the prototype; share closer to the prototype than to any such member. 20 seeds, 240 trials per cell.

240 runs and do not affect the results. The raw cue is slightly less informative in the equidistant design (89.2%), because members are closer together there (85% pairwise vs. about 74.5% in A and B).

3.16 Experiment 15c: Is there information the readout misses? (new in V7)

Decoder-level MI is a lower bound: a low value could mean that the nearest-neighbour readout misses member information hidden in the small deviations of converged outputs from the prototype. Throughout this experiment Z denotes the *binarised* output. The analyses form a hierarchy: the nearest-neighbour decoder tests conventional retrieval; a trained decoder tests whether one richer, family-relative readout extracts more; and an identical-output bound limits what *any* decoder could extract from Z in the sampled conditions (`conn_decoder_cv.py`; 40 new seeds, 480 trials per condition).

Trained decoder. For each candidate member, a family-relative feature vector is computed from the output: agreement with the member on all bits, on the member's own bits (where it differs from the family majority), on bits where other members deviate, and squared terms. Features are z-scored with the training fold's mean and standard deviation. A conditional-logit model with fixed L2 penalty 0.01 (chosen in advance, not tuned) is fitted by gradient descent (3000 steps) on half of the seeds and tested on the other half, in both directions, so no family is seen in training and testing. Held-out predictions from both folds are pooled before computing accuracy, MI and the seed-stratified permutation test. One decoder of this kind does not exhaust all possible decoders.

Decoder-independent bound. For each family and noise draw we run the network from all four possible cues with the same flipped positions and jitter. If the four outputs are identical, the output carries no information about which member was cued, for any decoder. Writing q for the share of such cases, $I(Y; Z \mid S) \le I(Y; Z \mid S, M) \le 2(1 - q)$ bits, where M is the noise draw (independent of Y), and no decoder can exceed an accuracy of $0.25q + (1 - q)$. The inequalities are exact for the true q; the table reports the observed q over 120 family/noise combinations, with 95% intervals from resampling the 40 seeds (5000 replicates). Because both bounds decrease in q, the upper limit of each bound's interval is obtained from the lower limit of the interval for q, and vice versa.

The trained decoder matches the nearest-neighbour readout everywhere: on the raw cue it recovers the same information, and on converged outputs it finds none. The bound is independent of any decoder.

Design	Condition	NN readout	Trained, held-out seeds	Identical outputs q [95% CI]	MI bound, bits [95% CI]	Accuracy bound
A: star	raw cue	94.0% / 1.61	93.5% / 1.58	0%	–	–
	converged continuous	25.8% / 0.00	26.2% / 0.00	87.5% [81.7, 93.3]	0.25 [0.13, 0.37]	34% [30, 39]
	converged synchronous	25.8% / 0.00	26.5% / 0.00	79.2% [70.8, 86.7]	0.42 [0.27, 0.58]	41% [35, 47]
B: hidden prototype	raw cue	96.5% / 1.74	96.2% / 1.71	0%	–	–
	converged continuous	26.0% / 0.00	25.6% / 0.00	79.2% [72.5, 85.8]	0.42 [0.28, 0.55]	41% [36, 46]
	converged synchronous	28.1% / 0.00	28.1% / 0.00	55.8% [46.7, 65.0]	0.88 [0.70, 1.07]	58% [51, 65]
C: equidistant members	raw cue	92.1% / 1.50	92.1% / 1.50	0%	–	–
	converged continuous	25.2% / 0.00	25.2% / 0.00	95.8% [92.5, 99.2]	0.08 [0.02, 0.15]	28% [26, 31]
	converged synchronous	25.6% / 0.02	25.6% / 0.02	92.5% [86.7, 97.5]	0.15 [0.05, 0.27]	31% [27, 35]

Table 13: Experiment 15c. Discrimination / decoder-level MI (bits; 0.00 = clipped, unclipped estimates −0.012 to −0.002 in `decoder_cv_results.csv`). Bounds apply to the binarised output. Seed-stratified permutation tests on the trained decoder: $p = 1.0$ for all converged conditions except synchronous C ($p = 0.07$); $p < 0.001$ for the raw cue.

Binarised converged outputs are the same whichever member was cued in 56–96% of sampled cases, so they carry an estimated 0.08–0.88 of the 2 bits (upper 95% limits 0.15–1.07). The bound is loose where outputs still vary (synchronous B), and there the trained decoder still finds nothing; where it is tight (design C) it rules out hidden member information directly. We therefore state the Hopfield result as: neither tested decoder detects member information, and the information in the binarised output is bounded below its maximum, tightly in design C and loosely in synchronous design B. The bound does not extend to the raw continuous state: converged continuous states are not identical across cues but differ by small residuals (median maximum deviation 2×10^{-6} in C, 2×10^{-4} in A, 5×10^{-3} in B; identical within 10^{-3} in 89%, 68% and 24% of cases), and we make no claim about information in those residuals.

4 A common trend across the tested systems

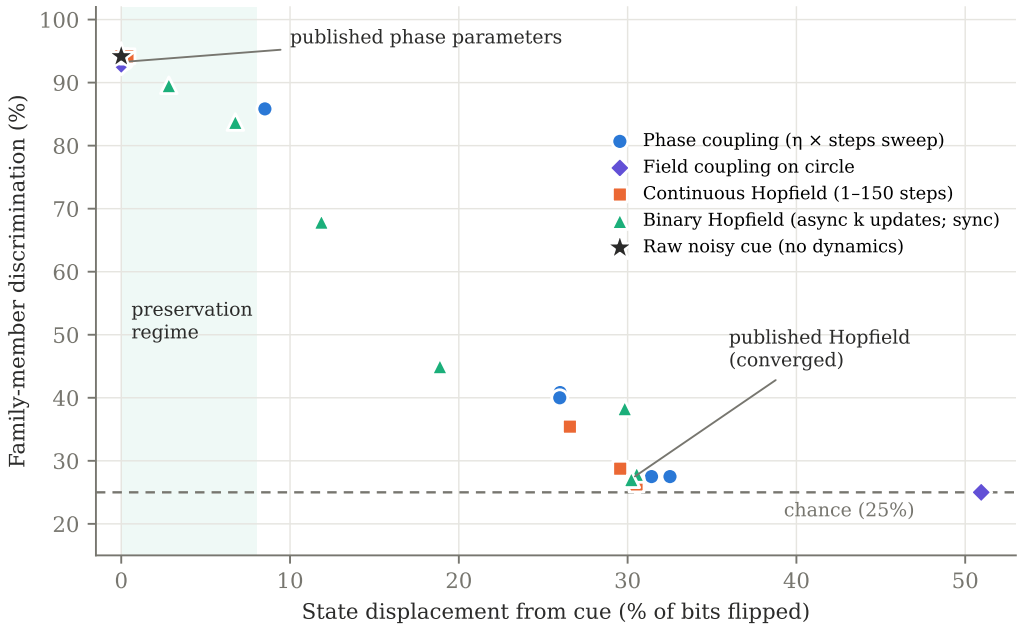


Figure 1: Family-member discrimination versus displacement of the output from the cue, for 31 configurations: the phase η×steps sweep (Experiment 14), centred field coupling, continuous Hopfield after 1–150 steps, and binary Hopfield with asynchronous or synchronous updates (Experiment 13). The tested configurations follow a common decreasing trend (Spearman $\rho = -0.95$, descriptive only). Shading: preservation regime (< 8% of bits moved). Data in `parameter_sweep_results.csv`, `matched_depth_results.csv` and `hopfield_depth_results.csv`.

Figure 1 summarises the paper. Across the couplings tested (phase differences, fields on the circle, Hebbian sums) and the nonlinearities tested (sine, tanh, sign), member discrimination follows a common decreasing trend with output displacement from the noisy cue (Spearman $\rho = -0.95$ over the 31 points; reported descriptively, as the points are not independent). The association is consistent with an integration-depth explanation, but it does not establish displacement as a universal predictor of associative-memory

performance: displacement is measured after binarisation, equal displacements can change different bits, and the 31 configurations come from parameter sweeps of a few systems and are not independent observations. Within the preservation regime every system keeps the 93–94% present in the cue. Beyond about 25% displacement every system is near chance. V4 attributed the difference between phase long (38.3%) and field long (25.0%) to a small structural advantage of pairwise coupling; Figure 1 shows that field long simply moved twice as many bits (50.9% vs. 26.1%), and at about 26% displacement phase (40%) and continuous Hopfield (35%) are close. We find no evidence for a coupling-structure effect once displacement is matched.

5 Mechanism

Converged Hebbian dynamics retrieve the prototype. For a family of patterns that agree on 85% of positions, the Hebbian field $h_i = \sum_j W_{ij} s_j$ is dominated by the shared components and points towards the family centroid. Dynamics that run to a fixed point therefore end at or near the prototype, erasing most of the positions that identify the member. When no prototype is stored, they converge onto the unstored prototype when members are equidistant, and to prototype-like states near it when members are farther apart (Table 12). This is the classical prototype or concept attractor [3, 4, 6].

Gentle dynamics keep what the cue contains. A noisy cue with 25% flipped bits still identifies the member 92.9–94.2% of the time by nearest-neighbour readout. Any dynamics that move the state little preserve this. At the published phase parameters the dynamics are effectively the identity map: the result does not depend on J, the coupling function, or the load.

What bit accuracy measures. A converged network that returns the prototype has bit accuracy close to the prototype-only baseline (Experiment 15) and rising with within-family similarity (Experiment 10), which is higher than the bit accuracy of a near-identity map applied to a 25%-noise cue (75%). Bit accuracy thus ranks prototype collapse above identity preservation. Decoder-level member information ranks them correctly.

6 Discussion

Contribution. That Hebbian networks form prototype attractors for correlated patterns is known [3, 4, 6, 7], as is the feature-to-prototype transition in dense associative memories [5]. The contribution here is methodological. (i) The bit-accuracy inversion is a concrete, quantified case in which the standard retrieval metric reports success for a network from whose output the tested decoders recover no member information, and whose binarised output is shown, by a decoder-independent bound, to carry little. (ii) The integration-depth analysis shows that apparent architectural advantages in this task can arise entirely from how long each system is run, and that comparisons between associative memories should match displacement from the cue. (iii) Decoder-level member information with proper bias correction and seed-clustered inference is a simple metric that separates the two regimes, and it is robust to how the correlated families are constructed (Experiment 15).

Relation to the original hypothesis. The study set out to show that phase-difference coupling discriminates correlated patterns better than field coupling. Under the controls reported here, it does not: compared at similar output displacement, the tested phase, field and Hopfield systems show similar member discrimination (Figure 1). Displacement serves here as an operational measure of how far the dynamics have moved the state; equal numbers of steps or updates in different systems are not equivalent. We report this as a negative result for the original hypothesis.

Retrieval without convergence. The preservation regime is not a working associative memory: at small displacement the network does little error correction, and with random initial states it retrieves nothing (24.6%). The interesting open question is whether any dynamics can both correct cue noise and keep member identity for correlated patterns. In our tested regime, modern Hopfield networks do so by effectively explicit lookup (Experiment 12); learning rules that decorrelate patterns, such as the pseudo-inverse rule [19] or unlearning [20], are the classical alternatives and were not tested here.

Biology. Hippocampal phase coding [21, 22] has been proposed as a substrate for fine discrimination. Our results do not support a special role for phase coupling in this task; they suggest instead that any circuit operating with brief, non-convergent dynamics would preserve within-category detail that full attractor settling erases. No biological data are presented.

7 Limitations

- **Scale and structure.** $N = 80$ dense (and $N = 1024$ sparse); 3 families of mostly 4 members; random binary patterns. Three family geometries were tested (Experiment 15); real correlated data (e.g. similar handwritten digits) were not.
- **Readout.** Member identity is read by an external nearest-neighbour step on the binarised output. This measures whether the information is present in the output, not whether the network computes it.
- **MI estimation.** 180–480 trials per condition; Miller–Madow is a first-order correction and can slightly exceed the 2-bit maximum at perfect accuracy (clipped); clipping at zero adds a small upward bias near the null. Decoder-level MI is a lower bound on the information in the output state. Experiment 15c tests one trained decoder and gives a decoder-independent upper bound; the bound is tight only where most outputs are identical across cues (designs A and C).
- **Depth matching.** Displacement is measured in flipped bits after binarisation; sub-threshold phase drift is not captured.
- **Untested alternatives.** Pseudo-inverse and unlearning rules, sparse distributed memory, and correlated-pattern learning rules.
- **No formal proof** of prototype collapse under the dynamics studied; the mechanism is supported empirically.
- **Earlier numbers.** Experiments 1–6, 11 and 12 are reported as in V4 (no MI involved); Experiments 7–10 and 14 were re-run with the corrected estimator. Seed-clustered intervals are reported for Experiments 7, 13 and 15; the MI values in Experiments 8–10 and 14 are point estimates.

8 Conclusion

For correlated patterns, converged Hebbian retrieval tends toward the family prototype or prototype-like states close to it. Bit-level recall then reports high accuracy (83–92% across three family designs) while neither the nearest-neighbour nor the trained decoder detects reproducible member information in the converged outputs, and decoder-independent empirical bounds restrict the information in the binarised outputs further, most tightly in the equidistant design, and a near-identity map with lower bit accuracy (75%) keeps 1.70 of 2 bits. That map preserves cue information; it does not correct errors. Across the tested phase, field and Hopfield dynamics, discrimination is strongly associated with how far the state moves from the cue. The phase-coupling advantage reported in V1–V4 is an integration-depth effect. The robust findings are the bit-accuracy inversion, the depth dependence, and the recommendation that associative-memory comparisons on correlated data report decoder-level member information with seed-clustered statistics and match displacement from the cue.

9 AI contribution statement

This work was conducted with several AI systems; theoretical judgments and final decisions are the author's. **Claude (Anthropic):** implemented Experiments 1–12 and the V3 diagnostics; identified the integration-depth sensitivity; in V5, identified and corrected the Miller–Madow sign error, re-ran Experiments 7–10 and 14, designed and ran Experiment 13 and produced Figure 1; in V6 and V7, implemented Experiments 15–15c and the seed-clustered statistics, and drafted V5–V8. **ChatGPT (OpenAI):** critical review; identified the noise-model asymmetry, requested the continuous Hopfield baseline, the component ablation and the coupling decomposition; in its review of V5, identified the star-shaped family geometry and requested the prototype-only baseline, hierarchical statistics, decoder-level terminology and the Gutfreund and Krotov–Hopfield references; in its reviews of V6 and V7, requested direct prototype-distance measurements, identified the conditional form of the data-processing bound, pointed out that a low decoder-level MI does not by itself bound the information in the output state, which led to Experiment 15c, and

requested seed-level uncertainty for the bounds, the restriction of the bound to binarised outputs, and the "bitwise accuracy" terminology. **Copilot (Microsoft):** post-publication review; independently identified the integration-depth issue and requested matched-depth baselines, the parameter sweep, bootstrap intervals and per-trial data; initial sparse implementation. **Gemini (Google):** found a bug in the sparse implementation; conceptual framing.

10 Code and data

All scripts use NumPy/SciPy and fixed seeds.

- Experiments 1–12: `conn_similarity.py`, `conn_similarity_ablation.py`, `conn_fair_comparison.py`, `conn_phase_decomposition.py`, `conn_final_validation.py`, `conn_sparse_validation.py`, `conn_modern_hopfield.py`, `conn_diagnostic_v3.py`.
- Mutual information, corrected estimator: `conn_mi_bootstrap.py` (Experiment 7), `conn_mi_pooled.py` (Experiments 8–10).
- Integration depth: `conn_parameter_sweep.py` (Experiment 14); `conn_hopfield_depth.py` (Experiment 13); `fig1_collapse.py` (Figure 1).
- New in V6: `conn_stats.py` (decoder-level MI, hierarchical bootstrap, seed-stratified permutation); `conn_family_geometry.py` (Experiment 15); `conn_prototype_distance.py` (Experiment 15b); `conn_decoder_cv.py` (Experiment 15c).
- Data: `per_trial_predictions.csv`, `parameter_sweep_results.csv`, `matched_depth_results.csv`, `hopfield_depth_results.csv`, `family_geometry_results.csv`, `prototype_distance_results.csv`, `decoder_cv_results.csv`.

A Changes from Version 8

1. **Permutation test for the trained decoder** described as a conditional test on the pooled held-out predictions (the decoder is not refitted under permutation).
2. **Bound intervals.** The methods state that the upper limit of each bound comes from the lower limit for q.
3. **Scope.** The abstract and discussion state that the decoder-independent bound applies to the binarised output, and that "no member information" refers to what the tested decoders detect.

B Changes from Version 7 to Version 8

1. **Terminology.** "Exact recall" is renamed *bitwise accuracy* and the "exact-recall inversion" the *bit-accuracy inversion*: the measure is per-bit agreement, not exact pattern retrieval.
2. **Scope of the decoder-independent bound.** The output Z in Experiment 15c is defined as the binarised output, and the bound is stated for it alone. Raw continuous Hopfield states are not identical across cues (residual differences of 10^{-6}–10^{-2}), so no bound is claimed for them.
3. **Uncertainty for the bounds.** Seed-level bootstrap intervals are reported for the identical-output fraction q and for both bounds; unclipped MI estimates are added to the data file.
4. **Decoder documentation.** Feature scaling, fixed regularisation, fold structure and the pooling of held-out predictions are stated.
5. **Displacement language.** Figure 1 and the Discussion describe a common decreasing trend at similar output displacement (Spearman $\rho = -0.95$, descriptive), not equivalence of architectures at matched "depth".
6. **Further reproducibility details**: continuous initial state, which state is returned on a two-cycle, the random-matrix control and the modern Hopfield implementation.
7. **Wording.** The five-test statement refers to nearest-neighbour readouts; the conclusion distinguishes tight from loose bounds; the Appendix C item on the held-out decoder is corrected.

C Changes from Version 6 to Version 7

1. **Prototype distances (Experiment 15b).** Direct Hamming distances replace the inference from aggregate metrics. Converged outputs coincide with the unstored prototype in the equidistant design but only approach it in the hidden-prototype design ("prototype-like retrieval").
2. **Withdrawn figure.** The V4/V5 statement that Hopfield converges to the prototype in 69.4% of trials and to mixture states in 30.6% is not reproduced by the deposited code. In Experiment 7, 96.7% of converged binary outputs are decoded as the prototype; Experiment 15b gives direct distances.
3. **Experiment 15c (new).** A trained decoder on held-out seeds detects no member information in converged outputs, and a decoder-independent bound (outputs identical for all four cues in 56–96% of cases) limits the information they carry to 0.08–0.88 bits. Claims about the Hopfield output are restated accordingly ("no recoverable member information" rather than "the information is lost").
4. **Reproducibility details** added: sign tie-breaking, asynchronous update order, two-cycle stopping rule, the role of the phase jitter, and how MI clipping is applied.
5. **Data-processing bound** stated conditionally on the stored family, $I(Y; D(Z, S)) \leq I(Y; Z \mid S)$.
6. **Preservation vs. retrieval.** Abstract and conclusion now state that the phase system preserves cue information and does not correct errors; the methods explain why an MI bootstrap interval excluding zero is not a test of dependence.

D Changes from Version 5 to Version 6

1. **New Experiment 15** on family geometry, with a prototype-only baseline and synchronous two-cycle counts. The bit-accuracy inversion holds in all three designs and is strongest when all members are equidistant.
2. **Seed-clustered statistics.** Hierarchical bootstrap and seed-stratified permutation tests replace trial-level ones in Experiments 7, 13 and 15. Experiment 7 intervals widen slightly (phase MI 95% CI [1.560, 1.881]; V5: [1.571, 1.846]).
3. **Binary Hopfield MI.** V5 described it as "marginal". It remains nominally significant in Experiment 7 ($p = 0.03$) but does not replicate in four other tests; it is now described as negligible.
4. **Terminology.** "Member-level MI" is now "decoder-level MI", with the data-processing lower-bound argument; the V5 remark that a held-out decoder would give a stricter estimate is superseded by the trained, seed-held-out decoder of Experiment 15c (V7).
5. **Scope of claims.** The displacement curve is stated as an empirical regularity of the tested systems, not a general law; the modern Hopfield result is restricted to the tested regime of saturated softmax.
6. **References.** Added Gutfreund (1988) and Krotov & Hopfield (2016). The title of [14] is restored to the current ResearchGate version ("Phase Coherence Constraints…"); the V5 change to "Topological Phase Constraints…" referred to the December 2025 version. The DOI previously printed in the header is removed pending verification of the ResearchGate records.

E Changes from Version 4 to Version 5

1. **Title and abstract** rewritten. The V4 title ("Pairwise Phase-Difference Coupling Preserves Fine-Grained Pattern Discrimination") and abstract findings (1), (5) and (8) stated a phase-coupling advantage that V4's own Section 3.15 showed to be an integration-depth effect.
2. **Mutual-information estimator corrected.** V4 computed $\hat{I}_{\text{plug-in}} + (m_{XY} - 1)/(2n)$: the sign was wrong (plug-in MI is biased upward and must be corrected downward), only the joint-cell count was used, and a correction in nats was added to bits. Experiments 8–10 in V4 also averaged uncorrected per-seed MI over 12 trials each, which inflates estimates further. V5 pools trials and applies the correction to each entropy term. Main changes: phase MI 1.726 → 1.703 bits; Hopfield MI 0.069–0.070 → 0.040–0.048 bits, now with a permutation test showing it is not, or only marginally, different from zero; phase MI at 95% similarity 1.266 → 0.84 bits.
3. **The "stronger result" remark** on MI above the uniform-error value is removed; uniform errors give the minimum MI for a given accuracy, so exceeding it is expected. The uniform-error value is 1.60 bits (V4: 1.56).

4. **New Experiment 13.** Hopfield networks stopped early preserve member identity. The V4 claims that sign() "destroys within-family discrimination in a single update step" and is "structurally incapable of gentle dynamics" are withdrawn: one synchronous step leaves 38.3%, and asynchronous or early-stopped dynamics preserve up to 94.2%.
5. **Coupling-structure effect withdrawn.** The V4 attribution of phase long (38.3%) vs. field long (25.0%) to pairwise coupling is replaced by the displacement explanation (Figure 1).
6. **Experiment 9, $\alpha = 0.650$:** the deposited script gives 96.1%; V4's 93.3% came from a separate run.
7. **Matched-depth table:** the Hopfield displacement was printed as 0.0% because the script did not measure it; it is $\approx 30\%$.
8. **Figure 1 (new)** replaces V4's Figure S1.
9. **Prior work:** added the classical prototype-attractor literature [3, 6], recent dense-memory work [9] and MI-bias references [15, 17, 16].
10. **References:** Ramsauer et al. cited as ICLR 2021; all references now cited in the text; Cook (1989) removed as unused.
11. **Typography:** the "$\lesssim$" symbol that rendered as a black box in V4 is fixed.

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